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Project Report

CS-E4880 Machine Learning in Bioinformatics

Multiple kernel learning for drug-protein binding affinity prediction

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Abstract

Drug-like chemical compounds execute their actions mainly by modulating cellular targets, such as proteins. The development of computational methods that could provide fast, large-scale and systematic pre-screening of chemical probes is therefore on rise. A proper representation and use of similarities, equivalent to a kernel choice, is prerequisite for the achievement of high-quality drug-protein interaction (DPI) predictions. Multiple kernel learning (MKL) methods enable to combine data from different sources simultaneously and therefore are of great importance in machine learning field. In this paper is described a method for predicting drug-protein binding affinities using multiple protein kernels as well as drug kernel in MKL regression framework.

1 Introduction

Drugs are substances used to diagnose, cure, treat, or prevent disease. Their mechanism of action is based on modulating cellular targets within the body, such as proteins. (Cichonska et al. 2015)

According to the Lipinski's rule of five, an orally-absorbed drug fulfills following criteria with at most one exception: it has no more than 5 hydrogen bond donors, no more than 10 hydrogen bond acceptors, a molecular weight lower than 500 daltons, and an octanol-water partition coefficient logP (a measure of lipophilicity) not greater than 5. (Cichonska. 2019) However, there is still between $10^{20} - 10^{24}$ molecules that have features consistent with good pharmacological properties. This is a huge motivation for computational methods in drug discovery, because having that many drug-like compounds, it is simply not feasible to determine all the possible drug-protein interactions in the laboratory.

Machine learning methods are often applied for *in silico* drug screening. The aim is to derive rules from the existing bioactivity data (a phase of learning from training data) in order to build predictive models that can be then applied to infer unmeasured drug-protein binding affinities (prediction phase). In this project the area of focus is on system-based methods, where available bioactivity data together with information about drugs and proteins are used for training models under the assumption that similar drugs are likely to interact with similar proteins. (Cichonska. 2019) The method involves having drug space (and knowledge of drug-drug connections), protein space (and knowledge of protein-protein connections), known drug-protein interaction (DPI) network and new DPI's prediction as shown in Fig. 1.

Similarities between molecules are encoded using kernel functions resulting in drugs and proteins kernels, which are used in classification or regression algorithm, based on which the DPIs are predicted.

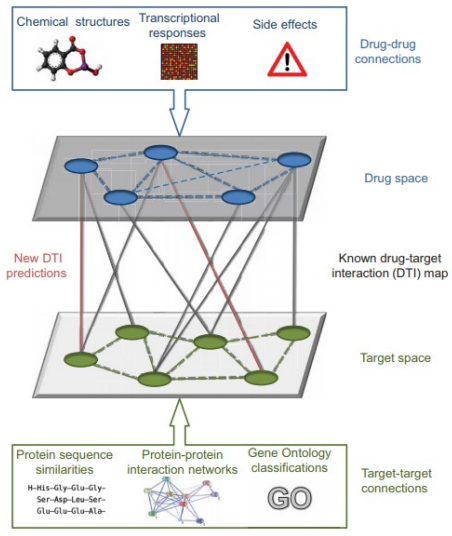

Fig. 1: The components of DPI prediction methods (Cichonska et al. 2015)

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2 Materials and methods

The data used in this study is a subset of data from experimental study by Metz *et al.* (2011). The dataset consists of 50 drug compounds and 50 protein targets together with their DPIs represented as pK_i's. pK_i corresponds to inhibition constant K_i in -log10 scale, where K_i is the drug compound concentration required for inhibition of protein's enzymatic activity by 50%.

For drug compounds, Tanimoto kernels based on several fingerprints are computed. Molecular fingerprints encode properties of small molecules and assess their similarities computationally through bit string comparisons (Fig. 2) that represent the presence or absence of particular substructers in the molecule. (Muegge et al. 2015) (Cichonska. 2019)

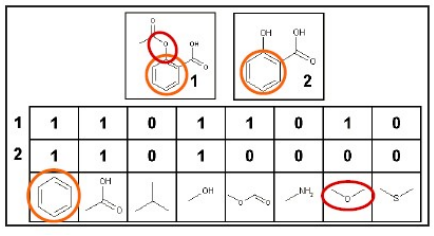

Fig. 2: Example of molecular fingerprint (Cichonska. 2019)

Fingerprint-based Tanimoto kernel is calculated as:

$$K(fp_1, fp_2) = \frac{N_{fp_1,fp_2}}{N_{fp_1} + N_{fp_2} - N_{fp_1,fp_2}} \tag{1}$$

where fp_i is a fingerprint of the molecule i, N_{fp_i} is number of 1-bits in the fingerprint fp_i and N_{fp_1,fp_2} is number of 1-bits in both fingerprints.

For proteins, normalized Smith-Waterman amino acid sequence alignment is used as one kernel together with several other Generic String kernels.

Smith-Waterman (SW) algorithm uses dynamic programming to perform local sequence alignment by comparing segments of all possible length. The optimal alignment is based on used scoring system (different scoring matrices, e.g. BLOSUM, PAM).

Normalized similarity between two proteins p_1 and p_2 is calculated in following way:

$$s(p_1, p_2) = \frac{SW(p_1, p_2)}{\sqrt{SW(p_1, p_1)}\sqrt{SW(p_2, p_2)}} \tag{2}$$

Generic String (GS) kernel is calculated as:

$$k_p(\mathbf{x_p}, \mathbf{x_p}') = \sum_{l=1}^{L} \sum_{i=0}^{|\mathbf{x_p}|-l} \sum_{j=0}^{|\mathbf{x'_p}|-l} exp\left(-\frac{(i-j)^2}{2\sigma_1^2}\right)$$

$$\exp\left(-\frac{\|\xi^l(x_{p_i+1},\ldots,x_{p_i+l}) - \xi^l(x'_{p_i+1},\ldots,x'_{p_i+l})\|^2}{2\sigma_2^2}\right) \tag{3}$$

Then, pairwise kernels that directly relate drug-protein pairs were constructed by taking a Kronecker product of each pair of drug kernel and protein kernel.

Kronecker product is calculated in following way:

$$\mathbf{K} = \mathbf{K_d} \otimes \mathbf{K_p} =$$

$$\begin{pmatrix} k_d(x_{d1}, x_{d1})\mathbf{K_p} & k_d(x_{d1}, x_{d2})\mathbf{K_p} & \ldots & k_d(x_{d1}, x_{d_{nd}})\mathbf{K_p} \\ k_d(x_{d2}, x_{d1})\mathbf{K_p} & k_d(x_{d2}, x_{d2})\mathbf{K_p} & \ldots & k_d(x_{d2}, x_{d_{nd}})\mathbf{K_p} \\ \vdots & \vdots & \ddots & \vdots \\ k_d(x_{d_{nd}}, x_{d1})\mathbf{K_p} & k_d(x_{d_{nd}}, x_{d2})\mathbf{K_p} & \ldots & k_d(x_{d_{nd}}, x_{d_{nd}})\mathbf{K_p} \end{pmatrix} \tag{4}$$

Those pairwise kernels are used with two-stage MKL algorithm ALIGNF to perform the binding affinity prediction. First, kernel mixture weights were determined based on maximising the centred alignment between the combined kernel ($\mathbf{K}_\mu$) and the response kernel derived from the label values ($\mathbf{K_y}$) in following way:

$$\underset{\mu}{\operatorname{argmax}}\, A(\mathbf{K}_\mu{}^c, \mathbf{K_y}) = max \frac{\langle \mathbf{K}_\mu{}^c, \mathbf{K_y} \rangle_F}{\|\mathbf{K}_\mu{}^c\|_F} \tag{5}$$

subject to: $\|\mu\|_2 = 1, \mu \geq 0$.

The optimization problem stated above can be solved by quadratic programming:

$$min \mathbf{v}^T \mathbf{M} \mathbf{v} - 2\mathbf{v}^T \mathbf{a} \tag{6}$$

$(\mathbf{a})_i = \langle \mathbf{K^{c(i)}}, \mathbf{K_y} \rangle_F$ i=1,...,L
$(\mathbf{M})_{ij} = \langle \mathbf{K^{c(i)}}, \mathbf{K^{c(j)}} \rangle_F$ i,j=1,...,L
where L is the number of pairwise kernels.

Considering the solution for above problem to be **v***, the optimal kernel weights are given by:

$$\mu* = \frac{\mathbf{v}*}{\|\mathbf{v}*\|} \tag{7}$$

The combined pairwise kernel is hence calculated as:

$$\mathbf{K}_\mu = \sum_{l=1}^{L} \mu_l \mathbf{K}^{(l)} \tag{8}$$

As a baseline model is used UNIMKL algorithm, which uses equal weights for each pairwise kernel:

$$\mu_l = \frac{1}{L} \tag{9}$$

In the second stage, combined kernel is used with Kernel Ridge Regression (KRR) as a prediction model.

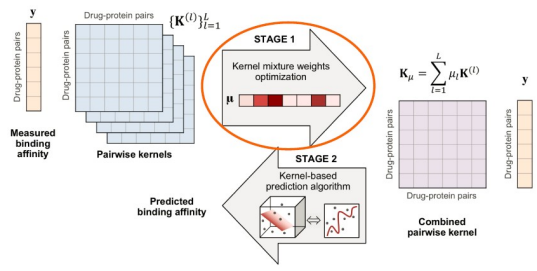

Fig. 3: Two stage MKL (Cichonska. 2019)

The solution to kernel ridge regression has a closed form:

$$\alpha = (\mathbf{K} + \lambda \mathbf{I}_N)^{-1} \mathbf{y} \tag{10}$$

The models' prediction can be represented in terms of inner products of training examples:

$$g(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x} \rangle = \sum_{i=1}^{N} \alpha_i \langle \mathbf{x}_i, \mathbf{x} \rangle = \sum_{i=1}^{N} \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \alpha^T \mathbf{k} \tag{11}$$

3 Results

Six different types of drug fingerprints (`standard`, `maccs`, `estate`, `pubchem`, `kr`, `circular`) from R package `rcdk` (Guha. 2017) were calculated and then used to compute fingerprint-based Tanimoto kernels.

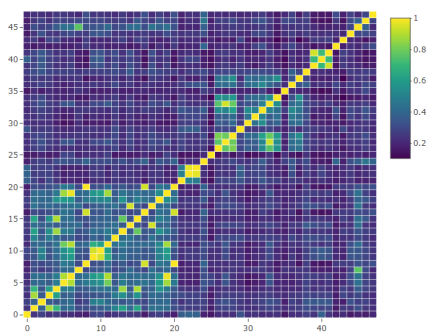

Fig. 4: Standard fingerprint-based Tanimoto drug kernel

Normalized Smith-Waterman (SW) similarity score based on pairwise aminoacids sequence alignments was computed and used as a protein kernel together with several Generic String (GS) kernels with different hyperparameters.

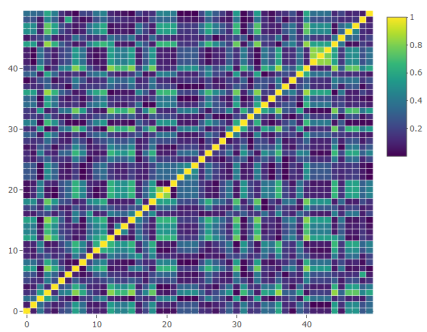

Fig. 5: Normalized Smith-Waterman (SW) protein kernel

As meaningful hyperparameters for GS kernels were considered those which resulted in values more spread all over the histogram. Use of substrings of a length equal to 10 for $\sigma = 1, 2, 2.5, 3, 3.5, 4, 5, 5.5, 6$ and substrings of a length 15 with $\sigma = 2, 4$ was evaluated as meaningful.

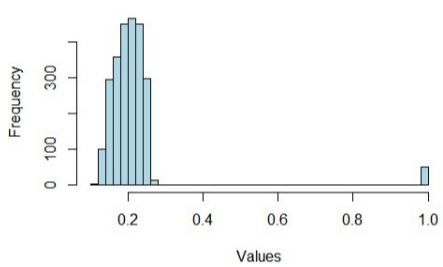

Fig. 6: Histogram of GS kernel values for $\sigma = 1$ (less meaningful)

Table 1. Models' performance

	RMSE		Pearson correlation	
	UNIMKL	**ALIGNF**	**UNIMKL**	**ALIGNF**
1	0.52998	0.83842	0.84145	0.71371
2	0.51336	0.99260	0.84542	0.64292
3	0.56932	1.29850	0.78534	0.25842
4	0.43687	0.96680	0.91766	0.69584
5	0.52086	0.93928	0.79217	0.60676
6	0.46791	0.73922	0.85864	0.76076
7	0.62297	0.93628	0.81765	0.76193
8	0.73535	1.17516	0.72030	0.45209
9	0.54873	0.93290	0.80036	0.54766
10	0.43587	0.80419	0.88461	0.65011

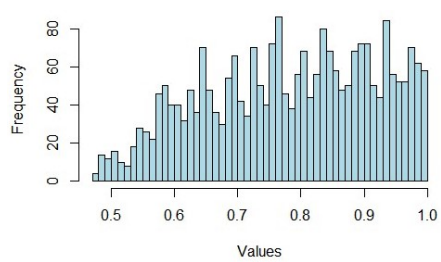

Fig. 7: Histogram of GS kernel values for $\sigma = 5$ (more meaningful)

In order to reduce computational cost were chosen only some of the GS kernels. As those were chosen ones with substrings of a length equal to 10 and $\sigma = 2, 2.5, 3, 3.5, 4, 5, 5.5$ and with substrings of a length 15 with $\sigma = 2, 4$. Also normalized SW protein kernel was used,

Pairwise kernels of 6 drug kernels and 10 protein kernels were constructed by taking a Kronecker product of each pair of drug kernel and protein kernel, resulting in 60 different pairwise kernels.

Two models, using UNIMKL and ALIGNF algorithms were computed. To tune the regularisation parameter λ, the nested cross validation was implemented with 10 outer folds and 3 inner folds. For all cases it resulted in $\lambda = 0.001$.

The performance was reported in the form of root mean squared error (RMSE) and Pearson correlation (Tab. 1). The mean RMSE for UNIMKL algorithm is and for ALIGNF algorithm is . The mean Pearson correlation for UNIMKL is and for ALIGNF is .

4 Conclusion

Several drug kernels as well as protein kernels were computed despite technical difficulties. 6 calculated drug kernels together with 10 protein kernels were used in MKL regression framework. Two prediction models, one using UNIMKL algorithm (where each pairwise kernel has equal weight in their combination to combined pairwise kernel) and other using ALIGNF algorithm (where kernel mixture weights are determined by maximisisng the centered alignment between the combined kernel and the response kernel) were computed. Based on evalution of performance by RMSE and Pearson correlation, it shows that the UNIMKL algorithm has better performance. However, this result seems to be inconclusive. One of the reason can be use of very small amount of data to reduce the computational complexity. Overall, the performance for both algorithms is not that good, as the RMSE is between 0.52998 and 1.2985 and the predicted values are in range from 5 to 9.

In order to be able to obtain better results, more input data has to be used, which unfortunately was not possible in this case.

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